

Ansible Multi-Tier Application Deployment Project

Project Introduction

This project demonstrates the deployment and configuration of a multi-tier application infrastructure using Ansible automation. The environment consists of a load balancer, web server, database server, and an Ansible control node. The objective of the project was to automate infrastructure provisioning, application deployment, service configuration, and security management using Ansible playbooks and Galaxy roles.

Infrastructure Overview

The infrastructure was built using four virtual machines.

Node	Role	Operating System
node1	Ansible Control Node	RHEL 10
node2	Load Balancer	RHEL 9
node3	Web Server	Ubuntu
node4	Database Server	RHEL 9

The infrastructure follows a standard multi-tier architecture where incoming requests are handled by the load balancer and forwarded to the backend web server, which communicates with the database server.

Step 1 — Project Structure and Roles

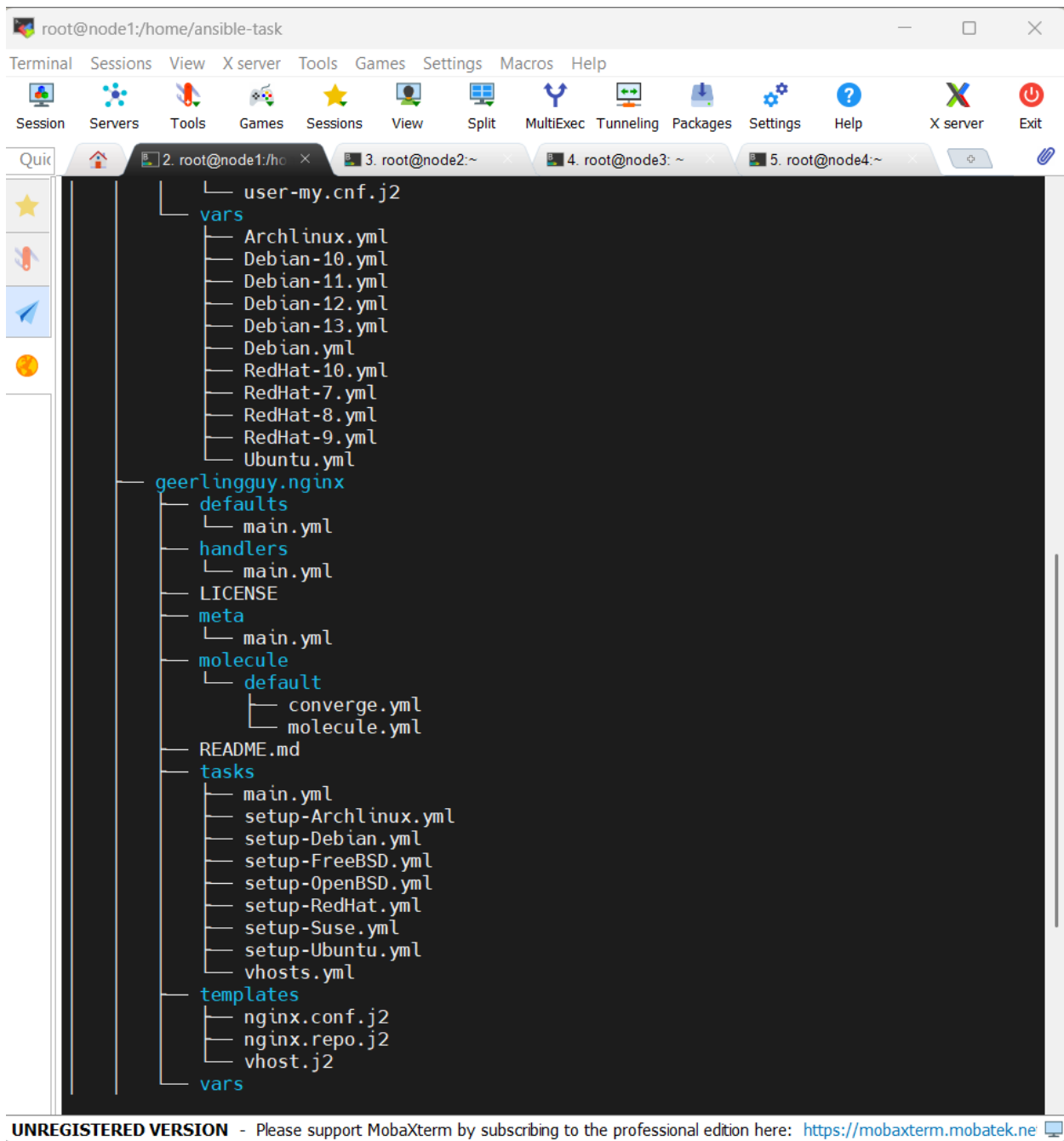
A dedicated Ansible project directory was created to organize playbooks, roles, templates, and variables. Separate playbooks were prepared for configuring the load balancer, web server, and database server.

Ansible Galaxy roles including:

- `geerlingguy.nginx`
- `geerlingguy.mysql`

were installed and integrated into the project to simplify infrastructure automation.

A custom security role was also created to apply basic SSH hardening configurations across systems.



```

├── vhost.j2
├── vars
│   ├── AlmaLinux.yml
│   ├── ArchLinux.yml
│   ├── Debian.yml
│   ├── FreeBSD.yml
│   ├── OpenBSD.yml
│   ├── RedHat.yml
│   ├── Rocky.yml
│   └── Suse.yml
├── security
│   └── tasks
│       └── main.yml
├── templates
│   └── nginx.conf.j2
├── web.log
└── web_servers.yml

```

25 directories, 69 files

[root@node1 ansible-task]#

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[root@node1 ansible-task]# ll

total 28

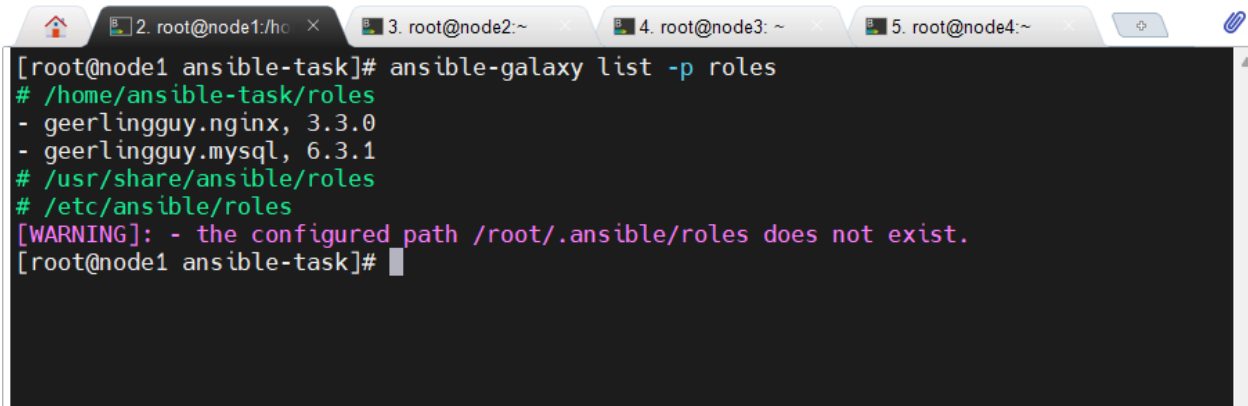
```

-rw-r--r--. 1 root root 73 May 10 21:32 ansible.cfg
-rw-r--r--. 1 root root 412 May 10 22:36 database.yml
drwxr-xr-x. 2 root root 51 May 10 22:25 group_vars
drwxr-xr-x. 2 root root 6 May 10 21:31 host_vars
-rw-r--r--. 1 root root 250 May 10 21:33 inventory.ini
-rw-r--r--. 1 root root 2651 May 10 22:52 lb.log
-rw-r--r--. 1 root root 499 May 10 21:53 load_balancer.yml
drwxr-xr-x. 5 root root 72 May 10 21:40 roles
drwxr-xr-x. 2 root root 27 May 10 21:51 templates
-rw-r--r--. 1 root root 841 May 10 22:53 web.log
-rw-r--r--. 1 root root 1104 May 10 22:17 web_servers.yml

```

[root@node1 ansible-task]#

Roles

A terminal window with a dark background and light-colored text. The window has four tabs at the top, labeled '2. root@node1:hc', '3. root@node2:~', '4. root@node3: ~', and '5. root@node4:~'. The active tab is '2. root@node1:hc'. The terminal shows the command 'ansible-galaxy list -p roles' being executed. The output lists two roles: 'geerlingguy.nginx, 3.3.0' and 'geerlingguy.mysql, 6.3.1'. It also shows the paths '/home/ansible-task/roles', '/usr/share/ansible/roles', and '/etc/ansible/roles'. A warning message is displayed: '[WARNING]: - the configured path /root/.ansible/roles does not exist.' The prompt '[root@node1 ansible-task]#' is visible at the bottom.

```
[root@node1 ansible-task]# ansible-galaxy list -p roles
# /home/ansible-task/roles
- geerlingguy.nginx, 3.3.0
- geerlingguy.mysql, 6.3.1
# /usr/share/ansible/roles
# /etc/ansible/roles
[WARNING]: - the configured path /root/.ansible/roles does not exist.
[root@node1 ansible-task]#
```

Step 2 — Load Balancer Configuration

The load balancer server was configured using Nginx. Ansible playbooks automated the installation and configuration of the Nginx service.

Backend web server IP addresses were defined using variables to support dynamic configuration.

Registers and conditions were implemented to monitor Nginx reload operations and detect failures during service reloads.

The load balancer successfully forwarded HTTP requests to the backend web server.

Inventory and loadbalancer files

```

[root@node1 ansible-task]# cat inventory.ini
[load_balancer]
lb1 ansible_host=192.168.13.128 ansible_user=root

[web_servers]
web1 ansible_host=192.168.13.134 ansible_user=root

[database]
db1 ansible_host=192.168.13.137 ansible_user=root

[all:vars]
ansible_python_interpreter=/usr/bin/python3
[root@node1 ansible-task]# cat load_balancer.yml
---
- name: Configure Load Balancer
  hosts: load_balancer
  become: yes

  roles:
    - geerlingguy.nginx
    - security

  tasks:
    - name: Deploy nginx configuration
      template:
        src: templates/nginx.conf.j2
        dest: /etc/nginx/nginx.conf

    - name: Reload nginx
      service:
        name: nginx
        state: reloaded
      register: nginx_reload

    - name: Notify if nginx reload failed
      debug:
        msg: "Nginx reload failed!"
        when: nginx_reload.failed
[root@node1 ansible-task]# █

```

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Step 3 — Web Server Deployment

The web server was configured on the Ubuntu server using Ansible automation. The deployment process included:

- Apache2 installation
- Git installation
- Application deployment
- Web service configuration

Blocks and rescue statements were implemented to support deployment error handling and rollback functionality.

The deployed application became accessible through the load balancer after successful configuration.

web_servers.yml

```
[root@node1 ansible-task]# cat web
web.log          web_servers.yml
[root@node1 ansible-task]# cat web_servers.yml
---
- name: Configure Web Server
  hosts: web_servers
  become: yes

  roles:
    - security

  tasks:

    - name: Install Apache and Git on Ubuntu
      apt:
        name:
          - apache2
          - git
        state: present
        update_cache: yes
        when: ansible_os_family == "Debian"

    - name: Start Apache
      service:
        name: apache2
        state: started
        enabled: yes
        when: ansible_os_family == "Debian"

    - block:

      - name: Clone application repository
        git:
          repo: "https://github.com/cloudacademy/static-website-example.git"
          dest: /var/www/html/app
          register: git_clone_result

      - name: Copy application homepage
        copy:
          src: /var/www/html/app/index.html
          dest: /var/www/html/index.html
```

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```
copy:
  src: /var/www/html/app/index.html
  dest: /var/www/html/index.html
  remote_src: yes

rescue:

  - name: Rollback deployment
    file:
      path: /var/www/html/app
      state: absent

  - name: Rollback message
    debug:
      msg: "Application deployment failed. Rollback completed."
[root@node1 ansible-task]#
```

Step 4 — Database Configuration

The database server was configured using the geerlingguy.mysql Ansible Galaxy role. MariaDB was installed, initialized, and configured securely.

Database credentials were stored using Ansible Vault encryption to protect sensitive information.

Database access was restricted specifically to the web server IP address, ensuring secure communication between application components.

User permissions and database privileges were configured successfully.

Database.yml ve **ansible vault**

```
[root@node1 ansible-task]# cat database.yml
---
- name: Configure Database Server
  hosts: database
  become: yes

  vars_files:
    - group_vars/database.yml

  vars:
    mysql_root_password_update: true
    mysql_root_use_socket: true

  pre_tasks:

    - name: Install MySQL Python package on RHEL
      dnf:
        name: python3-PyMySQL
        state: present
        when: ansible_os_family == "RedHat"

  roles:
    - geerlingguy.mysql
    - security
[root@node1 ansible-task]# cat group_vars/database.yml
$ANSIBLE_VAULT;1.1;AES256
62376166306431313961313930336139653335643262363361343239626136356636306462646136
3631656138376532643362336333643639643530313334390a643865653165646634323334663632
63306633613865323461376533363131373339633831326162376466333038616463663063616334
3165313361623038360a306636666138656666633338396330333866636634633631373566386363
36636465396436663936653561613536316339363264323063656235666430383664656330663430
62333838653365353139623532656532323561306135356433333262316436386534363965663964
65346337623165326638383031306136656437323333303231313934313233666231653433313334
31306530643934666438303033396237343234306161303363613933333038613865333639656331
3363636613266613961313132316231646633656261326162313463373334663734646438366136
39333666303239633534383763383338643130383564353937303463663930376161633737616634
34623732313431393030336339363335643538386139666331666231613838303435306365346262
63633435356266643761666137356432376262343861333463336463326134623837626135326531
3930
[root@node1 ansible-task]#
```

Security Implementation

Several security configurations were implemented during the project.

SSH hardening settings were applied through a custom security role, including:

- disabling empty passwords
- managing SSH configuration files

Database credentials were encrypted using Ansible Vault.

SELinux-related permission issues affecting MariaDB were identified and resolved by applying correct SELinux contexts and secure temporary directory configurations instead of disabling SELinux protections.

Troubleshooting and Problem Solving

During implementation, several infrastructure and configuration issues were encountered and resolved.

Repository configuration problems on RHEL systems were fixed using mounted local repositories.

SSH authentication problems caused by modified SSH settings were corrected by restoring proper SSH configuration values.

Nginx reload failures were diagnosed through configuration validation and service logs.

MariaDB startup failures related to SELinux permissions and temporary file creation were resolved using secure SELinux policy adjustments and directory permission corrections.

Verification and Testing

Deployment verification was performed by testing service availability and application accessibility across all infrastructure layers.

Successful Ansible connectivity to all servers was verified using Ansible ping modules.

Nginx, Apache2, and MariaDB services were confirmed running successfully.

The application was accessed through the load balancer using curl requests, verifying correct reverse proxy functionality.

Database creation, user permissions, and restricted host access were validated using MariaDB queries.

GitHub Repository

GitHub Repository Link:

<https://github.com/elvirosmanov1989-alt/ansible-multitier-project>

Conclusion

The project successfully automated the provisioning and configuration of a complete multi-tier application infrastructure using Ansible.

The implementation demonstrated practical experience in:

- infrastructure automation
- Linux administration
- service management
- secure credential handling
- load balancing
- database configuration
- troubleshooting real infrastructure issues

The project represents a practical DevOps automation workflow using Ansible and Linux systems.

```
[root@node1 ansible-task]# ansible all -m ping --ask-vault-pass
Vault password:
lb1 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
web1 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
db1 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
[root@node1 ansible-task]#
```

Node1: curl <http://192.168.13.128>

```
[root@node1 ansible-task]# curl http://192.168.13.128
```

```
<!DOCTYPE HTML>
```

```
<!--
```

```
    Dimension by HTML5 UP
```

```
    html5up.net | @ajlkn
```

```
    Free for personal and commercial use under the CCA 3.0 license (html5up.net/license)
```

```

-->
<html>
  <head>
    <title>Welcome</title>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1, user-scalable=no" />
    <link rel="stylesheet" href="assets/css/main.css" />
    <!--[if lte IE 9]><link rel="stylesheet" href="assets/css/ie9.css" /><![endif]-->
    <noscript><link rel="stylesheet" href="assets/css/noscript.css" /></noscript>
  </head>
  <body>

    <!-- Wrapper -->
    <div id="wrapper">

      <!-- Header -->
      <header id="header">
        <div class="logo">
          <span class="icon fa-diamond"></span>
        </div>
        <div class="content">
          <div class="inner">
            <h1>Dimension</h1>
            <p><!--[-->A fully responsive site template designed by <a
href="https://html5up.net">HTML5 UP</a> and released<!--[--><br />
            <!--[-->for free under the <a
href="https://html5up.net/license">Creative Commons</a> license.<!--[--></p>
          </div>
        </div>
        <nav>
          <ul>
            <li><a href="#intro">Intro</a></li>
            <li><a href="#work">Work</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#contact">Contact</a></li>
            <!--<li><a href="#elements">Elements</a></li>-->
          </ul>
        </nav>
      </header>

      <!-- Main -->
      <div id="main">

        <!-- Intro -->
        <article id="intro">
          <h2 class="major">Intro</h2>
          <span class="image main"></span>

```

<p>Aenean ornare velit lacus, ac varius enim ullamcorper eu. Proin aliquam facilisis ante interdum congue. Integer mollis, nisl amet convallis, porttitor magna ullamcorper, amet egestas mauris. Ut magna finibus nisi nec lacinia. Nam maximus erat id euismod egestas. By the way, check out my awesome work.</p>

<p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis dapibus rutrum facilisis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Etiam tristique libero eu nibh porttitor fermentum. Nullam venenatis erat id vehicula viverra. Nunc ultrices eros ut ultricies condimentum. Mauris risus lacus, blandit sit amet venenatis non, bibendum vitae dolor. Nunc lorem mauris, fringilla in aliquam at, euismod in lectus. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. In non lorem sit amet elit placerat maximus. Pellentesque aliquam maximus risus, vel sed vehicula.</p>

</article>

<!-- Work -->

<article id="work">

<h2 class="major">Work</h2>

<p>Adipiscing magna sed dolor elit. Praesent eleifend dignissim arcu, at eleifend sapien imperdiet ac. Aliquam erat volutpat. Praesent urna nisi, fringila lorem et vehicula lacinia quam. Integer sollicitudin mauris nec lorem luctus ultrices.</p>

<p>Nullam et orci eu lorem consequat tincidunt vivamus et sagittis libero. Mauris aliquet magna magna sed nunc rhoncus pharetra. Pellentesque condimentum sem. In efficitur ligula tate urna. Maecenas laoreet massa vel lacinia pellentesque lorem ipsum dolor. Nullam et orci eu lorem consequat tincidunt. Vivamus et sagittis libero. Mauris aliquet magna magna sed nunc rhoncus amet feugiat tempus.</p>

</article>

<!-- About -->

<article id="about">

<h2 class="major">About</h2>

<p>Lorem ipsum dolor sit amet, consectetur et adipiscing elit. Praesent eleifend dignissim arcu, at eleifend sapien imperdiet ac. Aliquam erat volutpat. Praesent urna nisi, fringila lorem et vehicula lacinia quam. Integer sollicitudin mauris nec lorem luctus ultrices. Aliquam libero et malesuada fames ac ante ipsum primis in faucibus. Cras viverra ligula sit amet ex mollis mattis lorem ipsum dolor sit amet.</p>

</article>

<!-- Contact -->

<article id="contact">

<h2 class="major">Contact</h2>

<form method="post" action="#">

<div class="field half first">

<label for="name">Name</label>

<input type="text" name="name" id="name" />

</div>

<div class="field half">

```

        <label for="email">Email</label>
        <input type="text" name="email" id="email" />
    </div>
    <div class="field">
        <label for="message">Message</label>
        <textarea name="message" id="message"
rows="4"></textarea>
    </div>
    <ul class="actions">
        <li><input type="submit" value="Send Message"
class="special" /></li>
        <li><input type="reset" value="Reset" /></li>
    </ul>
</form>
<ul class="icons">
    <li><a href="#" class="icon fa-twitter"><span
class="label">Twitter</span></a></li>
    <li><a href="#" class="icon fa-facebook"><span
class="label">Facebook</span></a></li>
    <li><a href="#" class="icon fa-instagram"><span
class="label">Instagram</span></a></li>
    <li><a href="#" class="icon fa-github"><span
class="label">GitHub</span></a></li>
</ul>
</article>

<!-- Elements -->
<article id="elements">
    <h2 class="major">Elements</h2>

    <section>
        <h3 class="major">Text</h3>
        <p>This is <b>bold</b> and this is <strong>strong</strong>. This
is <i>italic</i> and this is <em>emphasized</em>.
        This is <sup>superscript</sup> text and this is
        <sub>subscript</sub> text.
        This is <u>underlined</u> and this is code: <code>for (;) { ...
        }</code>. Finally, <a href="#">this is a link</a>.</p>
        <hr />
        <h2>Heading Level 2</h2>
        <h3>Heading Level 3</h3>
        <h4>Heading Level 4</h4>
        <h5>Heading Level 5</h5>
        <h6>Heading Level 6</h6>
        <hr />
        <h4>Blockquote</h4>
        <blockquote>Fringilla nisl. Donec accumsan interdum nisi, quis
tincidunt felis sagittis eget tempus euismod. Vestibulum ante ipsum primis in faucibus vestibulum. Blandit

```

adipiscing eu felis iaculis volutpat ac adipiscing accumsan faucibus. Vestibulum ante ipsum primis in faucibus lorem ipsum dolor sit amet nullam adipiscing eu felis.</blockquote>

<h4>Preformatted</h4>

<pre><code>i = 0;

```
while (!deck.isInOrder()) {  
    print 'Iteration ' + i;  
    deck.shuffle();  
    i++;  
}
```

```
print 'It took ' + i + ' iterations to sort the deck.';</code></pre>
```

</section>

<section>

<h3 class="major">Lists</h3>

<h4>Unordered</h4>

Dolor pulvinar etiam.

Sagittis adipiscing.

Felis enim feugiat.

<h4>Alternate</h4>

<ul class="alt">

Dolor pulvinar etiam.

Sagittis adipiscing.

Felis enim feugiat.

<h4>Ordered</h4>

Dolor pulvinar etiam.

Etiam vel felis viverra.

Felis enim feugiat.

Dolor pulvinar etiam.

Etiam vel felis lorem.

Felis enim et feugiat.

<h4>Icons</h4>

<ul class="icons">

Twitter

Facebook

Instagram

```

        <li><a href="#" class="icon fa-github"><span
class="label">Github</span></a></li>
    </ul>

    <h4>Actions</h4>
    <ul class="actions">
        <li><a href="#" class="button special">Default</a></li>
        <li><a href="#" class="button">Default</a></li>
    </ul>
    <ul class="actions vertical">
        <li><a href="#" class="button special">Default</a></li>
        <li><a href="#" class="button">Default</a></li>
    </ul>
</section>

<section>
    <h3 class="major">Table</h3>
    <h4>Default</h4>
    <div class="table-wrapper">
        <table>
            <thead>
                <tr>
                    <th>Name</th>
                    <th>Description</th>
                    <th>Price</th>
                </tr>
            </thead>
            <tbody>
                <tr>
                    <td>Item One</td>
                    <td>Ante turpis integer aliquet porttitor.</td>
                    <td>29.99</td>
                </tr>
                <tr>
                    <td>Item Two</td>
                    <td>Vis ac commodo adipiscing arcu aliquet.</td>
                    <td>19.99</td>
                </tr>
                <tr>
                    <td>Item Three</td>
                    <td>Morbi faucibus arcu accumsan lorem.</td>
                    <td>29.99</td>
                </tr>
                <tr>
                    <td>Item Four</td>
                    <td>Vitae integer tempus condimentum.</td>
                    <td>19.99</td>
                </tr>
            </tbody>
        </table>
    </div>

```

```
<td>Item Five</td>
<td>Ante turpis integer aliquet porttitor.</td>
<td>29.99</td>
</tr>
</tbody>
<tfoot>
<tr>
<td colspan="2"></td>
<td>100.00</td>
</tr>
</tfoot>
</table>
</div>
```

```
<h4>Alternate</h4>
<div class="table-wrapper">
  <table class="alt">
    <thead>
      <tr>
        <th>Name</th>
        <th>Description</th>
        <th>Price</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>Item One</td>
        <td>Ante turpis integer aliquet porttitor.</td>
        <td>29.99</td>
      </tr>
      <tr>
        <td>Item Two</td>
        <td>Vis ac commodo adipiscing arcu aliquet.</td>
        <td>19.99</td>
      </tr>
      <tr>
        <td>Item Three</td>
        <td>Morbi faucibus arcu accumsan lorem.</td>
        <td>29.99</td>
      </tr>
      <tr>
        <td>Item Four</td>
        <td>Vitae integer tempus condimentum.</td>
        <td>19.99</td>
      </tr>
      <tr>
        <td>Item Five</td>
        <td>Ante turpis integer aliquet porttitor.</td>
        <td>29.99</td>
```



```

        </tr>
      </tbody>
    </tfoot>
  </tr>
  <td colspan="2"></td>
  <td>100.00</td>
</tr>
</tfoot>
</table>
</div>
</section>

<section>
  <h3 class="major">Buttons</h3>
  <ul class="actions">
    <li><a href="#" class="button special">Special</a></li>
    <li><a href="#" class="button">Default</a></li>
  </ul>
  <ul class="actions">
    <li><a href="#" class="button">Default</a></li>
    <li><a href="#" class="button small">Small</a></li>
  </ul>
  <ul class="actions">
    <li><a href="#" class="button special icon fa-
download">Icon</a></li>
    <li><a href="#" class="button icon fa-
download">Icon</a></li>
  </ul>
  <ul class="actions">
    <li><span class="button special
disabled">Disabled</span></li>
    <li><span class="button disabled">Disabled</span></li>
  </ul>
</section>

<section>
  <h3 class="major">Form</h3>
  <form method="post" action="#">
    <div class="field half first">
      <label for="demo-name">Name</label>
      <input type="text" name="demo-name" id="demo-name"
value="" placeholder="Jane Doe" />
    </div>
    <div class="field half">
      <label for="demo-email">Email</label>
      <input type="email" name="demo-email" id="demo-
email" value="" placeholder="jane@untitled.tld" />
    </div>
    <div class="field">

```

```

        <label for="demo-category">Category</label>
        <div class="select-wrapper">
            <select name="demo-category" id="demo-category">
                <option value="">-</option>
                <option value="1">Manufacturing</option>
                <option value="1">Shipping</option>
                <option value="1">Administration</option>
                <option value="1">Human Resources</option>
            </select>
        </div>
    </div>
    <div class="field half first">
        <input type="radio" id="demo-priority-low"
name="demo-priority" checked>

        <label for="demo-priority-low">Low</label>
    </div>
    <div class="field half">
        <input type="radio" id="demo-priority-high"
name="demo-priority">

        <label for="demo-priority-high">High</label>
    </div>
    <div class="field half first">
        <input type="checkbox" id="demo-copy" name="demo-
copy">

        <label for="demo-copy">Email me a copy</label>
    </div>
    <div class="field half">
        <input type="checkbox" id="demo-human"
name="demo-human" checked>

        <label for="demo-human">Not a robot</label>
    </div>
    <div class="field">
        <label for="demo-message">Message</label>
        <textarea name="demo-message" id="demo-message"
placeholder="Enter your message" rows="6"></textarea>
    </div>
    <ul class="actions">
        <li><input type="submit" value="Send Message"
class="special" /></li>

        <li><input type="reset" value="Reset" /></li>
    </ul>
</form>
</section>

</article>

</div>

<!-- Footer -->

```

```
        <footer id="footer">
            <p class="copyright">&copy; Untitled. Design: <a
href="https://html5up.net">HTML5 UP</a>.</p>
        </footer>
```

```
    </div>
```

```
<!-- BG -->
```

```
    <div id="bg"></div>
```

```
<!-- Scripts -->
```

```
    <script src="assets/js/jquery.min.js"></script>
```

```
    <script src="assets/js/skel.min.js"></script>
```

```
    <script src="assets/js/util.js"></script>
```

```
    <script src="assets/js/main.js"></script>
```

```
</body>
```

```
</html>
```

```
[root@node1 ansible-task]#
```